

Kodu

Kodu is a programming language within a game. It allows players to create their own 3D games on Microsoft's Xbox 360 game console and Windows PCs.

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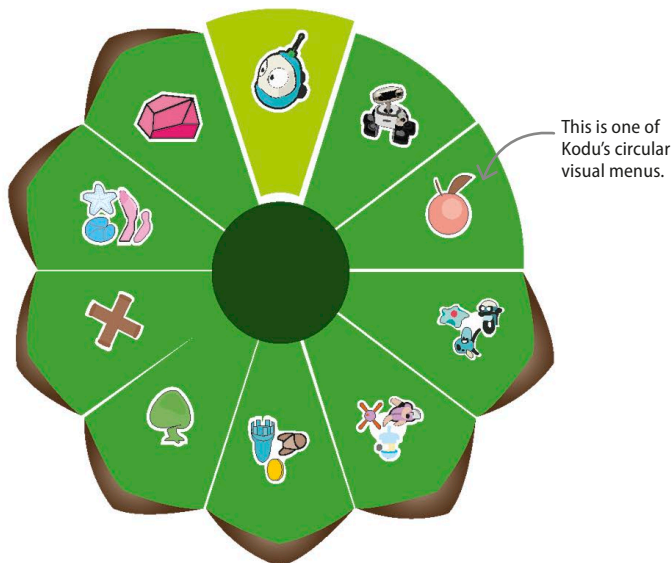
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Background

First released in 2009, Kodu is an application for Microsoft's Xbox 360 game console. As children in the 1980s, its developers enjoyed modifying video games by altering their code. Consequently, they felt that modern games deprived children of such opportunities, and decided to make a language that could be used to create a modern 3D video game world. Kodu aims to get kids to see coding as a creative tool for expressing their ideas.

▷ What type of programming?

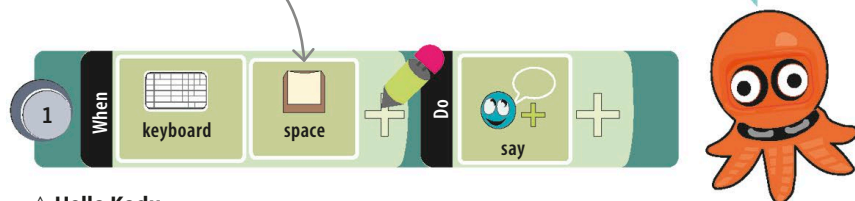
Kodu is a visual programming language. It's also object-oriented, as each character or item in the game world is an object with features that can be recognized or changed, and actions that can be done by or to them.



How does it work?

In Kodu, users write programs by creating new rules for the game world. The rules are made up of icons that represent items, actions, and properties, such as colour. The rules determine how the characters in the world react to various situations, and each is in the form, When: <condition> Do: <action>. Icons are selected from circular visual menus using a mouse or game controller.

The octopus says, "Hello world!" when the space key is pressed.



△ Hello Kodu

Kodu has a "say" command where users can type text that is displayed in speech or thought bubbles next to a character.

IN DEPTH

Built-in physics

Most game programmers have to write a lot of code to create the laws of physics that govern their game world. These laws usually mirror those in the real world. Kodu's developers decided that they would provide a game world with working physics, so that kids using Kodu can work on their own ideas without worrying about technicalities.

